

Course Plan

Research Methods (ARI-304)

Syllabus

General Information

Course Number	ARI-304
Credit Hours	3+0 (Theory Credit Hour = 3, Lab Credit Hours = 0)
Prerequisite	None
Batch & Semester	F23-AI (Semester VI (A & B)) – Spring 2026
University	Aror University Sukkur
Course Coordinator	Prof. Dr. Faheem Akhtar Rajput

Course Objectives

This course will prepare students for research projects by examining how to plan, conduct, and report on empirical investigations. The course will cover techniques applicable to each of the steps of a research project, including formulating research questions, theory building, data analysis (using both qualitative and quantitative methods), building evidence, assessing validity, and publishing.

It will particularly focus on research involving software, developing statistical/AI tools to measure software performance and the ways in which people interact with software tools.

In addition, this course focuses on developing research and writing skills in the Computer Science & Engineering domain. As software development requires a multidisciplinary approach, many of the concepts are borrowed from social science, psychology, statistics, and other domains. Here we investigate empirical research methods for their applicability and suitability to a research problem.

Mode of Delivery

Synchronous and Asynchronous mode of Lectures

- PDF Slides of every lecture will be uploaded on created class group after the commencement of lecture
- Interactive session will be conducted as per timetable

Assessment Criteria:

Interactive Mode of Class	
1. Class Participation/Activities	10%
2. Research Paper/Presentation	20%
3. Mid-Term Exam	30%
4. Final Exam	40%

Course Learning Outcomes

Course Learning Outcomes (CLOs)	
1	Research Problem Analysis and Investigation: Students will be able to analyze complex computing-related research problems, conduct systematic literature reviews, and identify gaps in knowledge using appropriate empirical and investigative methods from computer science and related disciplines.
2	Design and Development of Research Solutions: Students will be able to design, develop, and evaluate research methodologies, tools, and experiments to address software-related research problems
3	Ethical and Professional Research Communication: Students will be able to produce well-structured, ethically sound research proposals, papers, and presentations that communicate findings effectively to academic and professional audiences, adhering to publishing standards and research ethics.

CLO-GA Mapping

		SO IDs								
CLO ID	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10
CLO 1	0	0	1	0	0	0	0	0	0	0
CLO 2	0	0	0	1	0	0	0	0	0	0
CLO 3	0	0	0	0	0	0	0	0	1	0

Books

1. **Wohlin, C., Runeson, P., Host, M., Ohlsson, M. C., Regnell, B., & Wesslén, A.**
(2012). *Experimentation in software engineering*. Springer Science & Business Media.
2. **Booth, W. C., Booth, W. C., Colomb, G. G., Colomb, G. G., Williams, J. M., & Williams, J. M.**
(2008). *The craft of research*. University of Chicago press.
3. **Dawson, C. W.**
(2005). *Projects in computing and information systems: a student's guide*. Pearson Education.
4. **Olivier, M. S.**
(2009). *Information technology research: A practical guide for computer science and informatics*. Van Schaik.
5. **Creswell, J. W., & Creswell, J. D.**
(2017). *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications.

Course Content

Week No.	Topic
1-2	Introduction to Research
	Course Introduction
	What is Research?
	Value and Significance of Research
	Some Basic Terminologies used in Research
	Technical versus Review Paper
	Introduction to Academic Databases
	Conference versus Journal papers
	Publishers
	Impact Factors versus Non-Impact Factor Journals
	Good versus Bad Research
3-4	Classification of Research
	Classification of Research
	Classification according to Purpose, Use, Time, and Data Collection Techniques
	Class Activity on identifying types of research
	Introduction to Research Process
	Selecting research area and topic
	Conducting Literature Review
5-6	Conducting a Systematic Literature Review (SLR) to Identify the Research Gap
	Finding the Research Problem
	Solving the Research Problem
	Reporting the Research
	Review vs. Systematic Literature Review (SLR)
	Process of reviewing literature

	SLR guidelines and process
	Formulating keywords and query strings
	Searching quality academic databases
	Extracting and analyzing data
	Reporting the literature
7	Critical Appraisal of Research Article
	How to read a research paper
	Critical Appraisal of a paper
	Reading Exercise
	Critical Appraisal Exercise
8-9	Management Software: EndNote/Zotero, overleaf
	Working on Reference
	Finding and storing references
	Searching with keywords and operators
	Setting up custom groups
	Annotating PDFs
	Creating and editing citations
	Sharing research and accessing shared libraries
10 & 11	Writing the Research Proposal & Data Visualization
	Understanding research proposal structure
	Conceptualizing research
	Problem identification and justification
	Literature gap identification
	Writing problem statement, rationale, aims, research questions
	Writing literature review, research design, methodology
	Writing expected outcomes with timelines
12	Writing the Systematic Literature Review Paper
	Reporting the SLR
	Tips to write SLR
	Use of EndNote/Zotero software
	Adding and citing references
	Lab Activity on SLR
13 & 14	Quantitative/Qualitative/Experimental Research and Data Analysis
	Introduction to Quantitative Research and Research Design
	Introduction to Qualitative Research and Research Design
	Introduction to Experimental Research
	Discussions
15-16	Week 15 & 16: Final Student Presentations (Research Proposal & Paper)
	Group presentations
	Final Term Exam

Instructor Contact Details

For any course related queries registered students can contact via email / Course WhatsApp group [or](#) on LMS forum.

Graduate Attributes for BS in AI, MMG & Cyber Security

- 1) Academic Education
- 2) Knowledge for Solving Computing Problems
- 3) Problem Analysis
- 4) Design/Development of Solutions
- 5) Modern Tool Usage
- 6) Individual and Team Work
- 7) Communication
- 8) Computing Professionalism and Society
- 9) Ethics
- 10) Lifelong Learning

Approvals

Prepared By	Prof. Dr. Faheem Akhtar Rajput
Approved By	Not Specified
Last Update	2026/01/10